

senior product manager at Silicon Labs, MCUs with floating-point units (FPUs) have further compatibility for such applications and ease the computation process. An FPU-supported controller enables precision up to long decimal values. This gives a lot of flexibility in data calculation and eliminates the requirement of overflow-checks in the program.

MCUs with FPUs can be of great advantage in applications like location tracking, processing of measurement data from accelerometers and gyroimeters, or any application where data precision is quite important.

Designing the connectivity module. How would you want your devices to interact? At the moment, major wireless options include Wi-Fi, Bluetooth Low Energy (BLE), GSM and Zigbee. Designers need to pay attention to some essential criteria for choosing a connectivity module. Some of these factors include data throughput, connectivity range and speed required, power requirement, scalability, robustness and upgraded protocol. Designers must choose a technology that will survive the test of time.

Building an effective connectivity model depends on two components: transceiver of the module (which essentially consists of a receiver, a transmitter and an antenna) and network frequency.

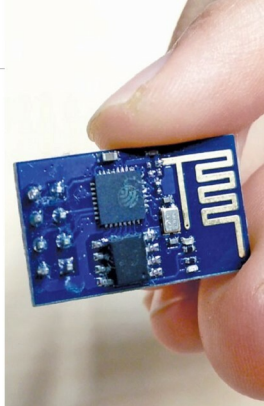
The receiver of a transceiver sets the threshold for the signal to be

received and also ensures that the signal is distinguishable from the other signals in the same frequency and the ambient noise. Therefore ensuring a high-quality receiver is imperative. Lower data rate improves sensitivity of the receiver.

The transmitter, on the other hand, controls the output power of the transceiver—how much it can amplify. Experts suggest a 6dB gain capacity can double the amplitude of the output signal. While higher transmitter power means higher output signal, there is a trade-off in battery life. Hence this decision should depend completely on the type of application, the data to be amplified and the geographic location.

Now let's talk about the frequency, which determines the range and data-throughput capacity. For IoT applications, sub-1.0GHz and 2.4GHz are the most used frequency ranges. Unlike the large data flow during media streaming or voice/video calls, IoT applications are associated with small data packets.

Now, many developers opt for 2.4GHz frequency range owing to its popularity in mainstream applications. This frequency range does provide higher data throughput capacity but not without challenges. First, the 2.4GHz array has a lower range of coverage and poor penetration through obstacles. More importantly, this frequency has a heavy traffic of data packets coming from



Wi-Fi IoT module (Image source: makerstream.blogspot.in)

other devices like microwave ovens and Wi-Fi devices. This creates the problem of signal interference.

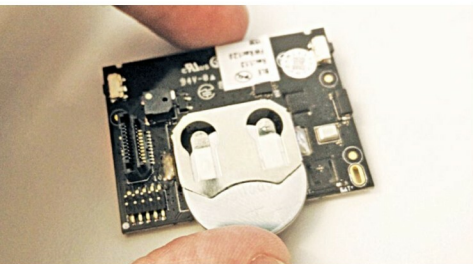
Sub-1.0GHz range, on the other hand, offers greater coverage range and sensitivity in lieu of data volume throughput. Distributed infrastructure can greatly benefit from its long-range characteristics. Moreover, data transmission through obstacles is possible.

Char suggests, "Since sensors of an IoT system generate small data throughout the day, sub-1.0GHz frequency is suited for most applications as it consumes much less power and provides connectivity for years on a battery."

For instance, IoT systems in agricultural fields or oil and gas industries will share very low-volume data such as soil pH, moisture level and gas level. A sub-1.0GHz frequency can efficiently cover the complete perimeter while consuming much less power.

Coming to wireless standards, while 802.11ac is the most popular Wi-Fi standard in the market, a newer iteration called IEEE 802.15.4 is gaining prominence for the IoT.

Creating a power-efficient system. For IoT applications, designers need to focus on power require-



Coin-cell battery-driven sensor module (Image source: thingsquare.com)